

PINK REPORT

Retrospective post-radical prostatectomy cohorts and systematic reviews generally suggest **no clear increase in biochemical recurrence (BCR)** among carefully selected men who start TRT after achieving/maintaining an undetectable PSA, though **prospective long-term oncologic safety data are limited**. In parallel, large randomized safety data in hypogonadal men without high prostate-cancer risk did **not show increased high-grade prostate cancer** with TRT, but this does not directly answer survivorship-specific risk. (**Gibson; Santucci; Kardoust Parizi; Flores; Pastuszak; Sarkar; Bhasin**)

Recommendation (evidence-guided): For a symptomatic hypogonadal man **12 months post-radical prostatectomy with favorable pathology and persistently undetectable PSA**, TRT **may be cautiously considered** using shared decision-making and **close PSA/testosterone/hematocrit monitoring**, ideally coordinated with urology/oncology as described in survivorship/urology guidance. (**NCCN; Mulhall; Bhasin [Endocrine Society]**)

References supporting summary ordered by strength of evidence:

Rank	Level of Evidence	Study Title	Authors	Journal (Year)	Notes
■ 1	Guideline	<i>Survivorship</i>	National Comprehensive Cancer Network (NCCN)	NCCN (Updated 2026)	Survivorship guidance acknowledges TRT may be considered in selected survivors and emphasizes coordination with treating team and ongoing surveillance.
■ 1	Guideline	<i>Evaluation and Management of Testosterone Deficiency: AUA Guideline</i>	Mulhall JP, Trost LW, Brannigan RE, et al.	<i>J Urol</i> (2018)	States TRT can be considered after radical prostatectomy in men with favorable pathology and undetectable PSA , with monitoring.
■ 1	Guideline	<i>Testosterone Therapy in Men With Hypogonadism: An Endocrine Society Clinical Practice Guideline</i>	Bhasin S, Brito JP, Cunningham GR, et al.	<i>J Clin Endocrinol Metab</i> (2018)	Notes limited/underpowered evidence for prostate outcomes; highlights uncertainty in quantifying risk-benefit and need for monitoring.
● 2	Clinical Trial	<i>Oncological Safety of Testosterone Replacement Therapy</i>	Santucci J, Stapleton P, Perera M, et al.	<i>BJU Int</i> (2025)	Systematic review of observational data; overall signals generally

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	Meta-analysis	<i>in Men With Localised Prostate Cancer: A Systematic Review of Observational Studies</i>			do not show increased recurrence, but evidence quality limited by design/follow-up.
● 2	Clinical Trial Meta-analysis	<i>in Prostate Cancer Survivors After Definitive Local Therapy: A Systematic Literature Review and Meta-Analysis</i>	Kardoust Parizi M, Abufaraj M, Fajkovic H, et al.	<i>Urol Oncol</i> (2019)	Meta-analysis across definitive local therapies; typically reports no increased recurrence signal; heterogeneous studies.
● 2	Clinical Trial Meta-analysis	<i>Testosterone Therapy Does Not Increase the Risks of Prostate Cancer Recurrence or Death After Definitive Treatment for Localized Disease</i>	Sarkar RR, Patel SH, Parsons JK, et al.	<i>Prostate Cancer Prostatic Dis</i> (2020)	Systematic review–type synthesis/reporting suggesting no increased recurrence/death; observational evidence base.
● 2	Clinical Trial Meta-analysis	<i>Testosterone Replacement Therapy Following Definitive Treatment for Prostate Cancer: A Scoping Review of Safety and Efficacy</i>	Gibson J, George M, Grice P, et al.	<i>Int J Impot Res</i> (2025)	Scoping review summarizing safety/efficacy; recurrence rates in post-prostatectomy TRT cohorts generally low but limited by retrospective designs.
● 2	Clinical Trial Meta-analysis	<i>Testosterone Treatment in Middle-Aged and Older Men with Hypogonadism</i>	Bhasin S, Snyder PJ.	<i>N Engl J Med</i> (2025)	Narrative review contextualizing TRT benefits/risks; not specific to post-prostate cancer survivors (useful framing, not direct survivorship evidence).
● 2	Clinical Trial Meta-analysis	<i>Testosterone Therapy in Men With Prostate Cancer</i>	Kaplan AL, Hu JC, Morgentaler A, et al.	<i>Eur Urol</i> (2016)	Review and recommendations based on available observational evidence/experience; supports careful selection/monitoring.

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 2	Clinical Trial Meta-analysis	<i>Testosterone Therapy in Men With Prostate Cancer: Literature Review, Clinical Experience, and Recommendations</i>	Morgentaler A, Connors WP.	<i>Asian J Androl</i> (2015)	Narrative review + clinical recommendations; emphasizes selection criteria and close PSA surveillance.
  3	Phase 3 (P3) Trial	<i>Prostate Safety Events During Testosterone Replacement Therapy in Men With Hypogonadism: A Randomized Clinical Trial</i>	Bhasin S, Travison TG, Pencina KM, et al.	<i>JAMA Netw Open</i> (2023)	Large RCT (TRAVERSE-related prostate safety outcomes): no increase in high-grade prostate cancer/prostate safety endpoints; excluded high-risk prostate cancer populations—indirect for survivorship counseling.
 4	Phase 2 (P2) Trial				No eligible P2 TRT trials in post-radical prostatectomy prostate cancer survivors cited in provided references.
 5	Phase 1 (P1) Trial				No eligible P1 TRT trials in post-radical prostatectomy prostate cancer survivors cited in provided references.
  6	Large Cohort Study	<i>Testosterone Therapy in Men After Radical Prostatectomy for Low-Intermediate Organ-Confining Prostate Cancer</i>	Flores JM, Vertosick EA, Salter CA, et al.	<i>J Urol</i> (2025)	Large single-institution cohort (incl. Grade Groups 1–3); TRT exposure not associated with increased BCR vs controls over available follow-up.
 7	Cohort Study	<i>Testosterone Replacement Therapy in Patients With Prostate Cancer After Radical Prostatectomy</i>	Pastuszak AW, Pearlman AM, Lai WS, et al.	<i>J Urol</i> (2013)	Retrospective cohort suggesting low recurrence rates with TRT in selected post-prostatectomy men; limited size/follow-up.
 7	Cohort Study	<i>Oncological Safety and Functional Outcomes of Testosterone</i>	Shahine H, Zanaty M, Zakaria AS, et al.	<i>World J Urol</i> (2021)	Post-RARP cohort reporting functional outcomes and oncologic surveillance;

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		<i>Replacement Therapy in Symptomatic Adult-Onset Hypogonadal Prostate Cancer Patients Following Robot-Assisted Radical Prostatectomy</i>			observational limitations apply.
7	Cohort Study	<i>Testosterone Replacement Therapy in Prostate Cancer Survivors Treated With Radiation With and Without Androgen Deprivation Therapy: A Retrospective Study and Narrative Review</i>	Chandra MS, Wang C, Gagliano-Jucá T, et al.	<i>Endocrine Practice</i> (2026)	Focused on post-radiation survivors (not prostatectomy-specific); provides symptomatic/outcome context; still retrospective.
8	Animal Model				No animal-model studies cited in provided references.